

I. ADAPTIVE GRAPH-BASED RECOMMENDATION ENGINE

A. Question Graph Construction

The question bank is modelled as a directed weighted graph $G = (V, E)$. Each node $v \in V$ carries attributes (subject, topic, ELO_v , $marks_v$, dur_v). Three semantically distinct edge types encode pedagogical structure:

- **Same-topic** (e_s): intra-topic edges ordered by increasing difficulty.
- **Subject-flow** (e_f): inter-topic edges within a subject following prerequisite order (e.g. *Linear Algebra Basics* $\rightarrow$ *Eigenvalues*).
- **Subject-bridge** (e_b): cross-subject edges encoding conceptual dependency (e.g. *Probability* $\leftrightarrow$ *Hypothesis Testing*).

Edge weights are a convex combination of four normalised distances:

$$w(u, v) = \alpha_1 d_{ELO} + \alpha_2 d_{marks} + \alpha_3 d_{topic} + \alpha_4 d_{type}, \quad \sum_i \alpha_i = 1 \quad (1)$$

where each $d_* \in [0, 1]$ (lower = more similar).

B. Learner State Representation

At each adaptive step the learner is described by the state tuple

$$s = (ELO, m, W, Q_{ans}, Q_{srv}, \pi_{rem}) \quad (2)$$

where $ELO \in [0, 3000]$; $m \in \{hot, steady, cold\}$ is the momentum label; $W \subseteq \mathcal{T}$ is the weak-topic set (accuracy $< \tau_w = 0.60$); Q_{ans} and Q_{srv} are the answered and already-served question sets (both excluded from future recommendations within the session); and π_{rem} is remediation progress metadata.

C. Momentum Computation

Momentum is derived from the learner's recent correctness streak k_{streak} and session accuracy $\hat{a}$:

$$m = \begin{cases} hot & k_{streak} \geq 5 \wedge \hat{a} \geq 0.80 \\ cold & k_{streak} \leq -3 \wedge \hat{a} < 0.50 \\ steady & otherwise \end{cases} \quad (3)$$

D. Target Policy

The engine shifts its ELO target based on momentum:

$$ELO_{tgt} = ELO + \Delta(m, \hat{a}), \quad \Delta = \begin{cases} +100 & m = hot, \hat{a} \geq 0.80 \\ +50 & m = steady, \hat{a} \geq 0.70 \\ 0 & otherwise \\ -100 & m = cold, \hat{a} < 0.50 \end{cases} \quad (4)$$

ELO_{tgt} is then mapped to a difficulty tier:

$$diff_{tgt} = \begin{cases} Easy & ELO_{tgt} < 1600 \\ Medium & 1600 \leq ELO_{tgt} < 2000 \\ Hard & 2000 \leq ELO_{tgt} < 2500 \\ Expert & ELO_{tgt} \geq 2500 \end{cases} \quad (5)$$

E. Neighbourhood Selection

Two neighbourhood modes are defined on G :

a) *Standard neighbourhood (non-remediation)*:

$$\mathcal{N}(q, k) = \{q' \in V : d_G(q, q') \leq k, q' \notin Q_{ans}, q' \notin Q_{srv}\} \quad (6)$$

with default hop limit $k = 2$.

b) *Remediation neighbourhood*: When an unresolved miss exists for question q_m :

$$\mathcal{N}_r(q_m, h) = \{q' \in V : d_G(q_m, q') \leq h, q' \notin Q_{ans}, diff(q') < diff(q_m)\} \quad (7)$$

with $h \in \{2, 3\}$. This guarantees candidates are related to the missed question (bounded hops) yet strictly easier (confidence recovery).

F. Candidate Scoring Function

Every candidate $q' \in \mathcal{C}$ receives a composite score:

$$score(q') = w_h \cdot h(q') + w_g \cdot g(q') \quad (8)$$

1) *Heuristic component* $h(q')$:

$$h(q') = 1 - \frac{w_1 d_{diff}(q') + w_2 d_{ELO}(q')}{C_{norm}} + \beta_w \mathbb{I}[q' \in W] + \beta_t \mathbb{I}[type(q') \neq type(q)] \quad (9)$$

where $d_{diff}(q') = |diff(q') - diff_{tgt}|$, $d_{ELO}(q') = |ELO_{q'} - ELO_{tgt}|$, β_w is a weak-topic bonus, and β_t is a type-diversity bonus.

2) *Graph-structural boost* $g(q')$:

$$g(q') = \gamma_1 \rho_{in}(q') + \gamma_2 \rho_{out}(q') + \gamma_3 c(q') \quad (10)$$

where $\rho_{in}, \rho_{out} \in [0, 1]$ are normalised in/out-degrees and $c(q')$ is normalised betweenness centrality. High-centrality nodes are *hub questions* at pedagogically critical junctions of the graph.

G. Full Recommendation Algorithm

II. ELO UPDATE RULE

After each answer submission the learner's ELO is updated using the standard Elo formula adapted for binary correctness:

$$E = \frac{1}{1 + 10^{(ELO_q - ELO)/400}} \quad (11)$$

$$ELO' = ELO + K(S - E) \quad (12)$$

where $K = 32$ is the update factor, $S \in \{0, 1\}$ is the binary outcome, ELO_q is the question's difficulty ELO, and E is the expected probability of a correct answer.

III. REMEDIATION AND RETRY ENGINE

A. Workflow

B. Exit Criterion

Remediation terminates when:

$$n_{steps} \geq N_r \wedge acc_{rem} \geq A_r \quad (N_r = 3, A_r = 0.75) \quad (13)$$

Algorithm 1 Adaptive Question Recommendation

Require: State s , graph G , hop limit $k=2$, remed. limit $h=3$ **Ensure:** Next question q^* with explanation string

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1: Compute  $m$  via Eq. (3)
2: Compute  $\text{ELO}_{\text{tgt}}$  via Eq. (4)
3: Compute  $\text{diff}_{\text{tgt}}$  via Eq. (5)
4: if  $\pi_{\text{rem}}.\text{active}$  then
5:    $C \leftarrow \mathcal{N}_r(\pi_{\text{rem}}.q_m, h)$  ▷ Eq. (7)
6: else
7:    $C \leftarrow \mathcal{N}(q_{\text{last}}, k)$  ▷ Eq. (6)
8: end if
9: if  $C = \emptyset$  then
10:   $C \leftarrow$  fallback: unseen questions nearest  $\text{diff}_{\text{tgt}}$ 
11: end if
12: for all  $q' \in C$  do
13:   compute  $h(q')$  via Eq. (9)
14:   compute  $g(q')$  via Eq. (10)
15:    $\text{score}(q') \leftarrow w_h h(q') + w_g g(q')$  ▷ Eq. (8)
16: end for
17:  $q^* \leftarrow \arg \max_{q' \in C} \text{score}(q')$ 
18: explain( $q^*$ ): emit hop distance, neighbourhood type,
    $\text{ELO}_{\text{tgt}}$ , bonuses applied
19: return  $q^*$ 

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Algorithm 2 Remediation and Retry

Require: Missed question q_m , hop limit h , thresholds $N_r=3$, $A_r=0.75$

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1: Mark  $q_m$  as UNRESOLVEDMISS; set  $\pi_{\text{rem}}$ 
2: repeat
3:    $q' \leftarrow$  recommendation from  $\mathcal{N}_r(q_m, h)$  via Alg. 1
4:   Serve  $q'$ ; collect answer; update  $\text{acc}_{\text{rem}}, n_{\text{steps}}$ 
5: until  $n_{\text{steps}} \geq N_r \wedge \text{acc}_{\text{rem}} \geq A_r$  ▷ Eq. (13)
6: Queue  $q_m$  for retry
7: if retry answer is correct then
8:   Clear UNRESOLVEDMISS; update ELO normally
9: else
10:  Re-activate  $\pi_{\text{rem}}$ ; carry miss to next session
11: end if

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IV. RAPID-GUESS DETECTION ENGINE

A. Minimum Solve-Time Estimate

A per-question minimum solve time guards against answer-velocity anomalies:

$$t_{\min}(q) = \alpha_0 + \alpha_1 \ell_{\text{stem}}(q) + \alpha_2 \ell_{\text{opts}}(q) + \alpha_3 c_{\text{idx}}(q) \quad (14)$$

with empirically set coefficients $\alpha_0=10$ s (baseline reading), $\alpha_1=0.02$ s/char, $\alpha_2=0.01$ s/char, $\alpha_3=30$ s for high-complexity items ($c_{\text{idx}} \in \{0, 1\}$).

B. ELO Penalty

If $t_{\text{spent}}(q) < t_{\min}(q)$, a proportional ELO penalty is applied to the standard gain g_{std} :

$$\text{ELO_gain} = \max\left(0, g_{\text{std}} - \left(1 - \frac{t_{\text{spent}}}{t_{\min}}\right) \cdot 20\right) \quad (15)$$

Correct-answer credit is preserved; only the ELO inflation is suppressed.

V. GATE SCORE PREDICTION MODEL

A. Unlock Gate

The predictor is locked until all four conditions hold simultaneously:

$$\text{UNLOCKED} \Leftrightarrow \begin{cases} \text{ELO} \geq 2500 \\ \text{comp}_{\text{mock}} \geq 0.70 \\ \text{comp}_{\text{topic}} \geq 0.70 \\ \text{comp}_{\text{adaptive}} \geq 0.70 \end{cases} \quad (16)$$

B. Component Normalisations

a) Accuracy (Acc_n):

$$\text{Acc}_n = \min\left(1, \frac{\text{acc}_{\text{raw}} - 0.50}{0.35}\right) \quad (17)$$

Maps raw accuracy to $[0, 1]$: 50% (above-guess floor) $\rightarrow 0$; 85% (strong) $\rightarrow 1$.

b) ELO (ELO_n):

$$\text{ELO}_n = \frac{\text{ELO} - 1600}{1200} \quad (18)$$

Maps baseline ELO 1600 $\rightarrow 0$ and ceiling 2800 $\rightarrow 1$.

c) Subject Consistency (Cons_n):

$$\text{Cons}_n = \max\left(0, 1 - \frac{\sigma(\{\text{acc}_i\}_{i=1}^n)}{\text{acc} + \varepsilon}\right) \quad (19)$$

where σ is standard deviation across subjects, acc is mean subject accuracy, and $\varepsilon = 0.01$ avoids division by zero. A learner with uniform 75% across all subjects scores $\text{Cons}_n = 1$; highly skewed subject coverage reduces the score.

d) Improvement Trend (Impr_n):

$$\Delta \text{acc} = \text{acc}_{\text{recent}} - \text{acc}_{\text{prior}} \in [-0.15, +0.15] \text{ (clipped)} \quad (20)$$

$$\text{Impr}_n = \frac{\Delta \text{acc} + 0.15}{0.30} \quad (21)$$

Maps -15% decline $\rightarrow 0$, stable $\rightarrow 0.5$, $+15\%$ improvement $\rightarrow 1$.

C. Score Basis

$$\text{SB} = 0.40 \text{Acc}_n + 0.30 \text{ELO}_n + 0.20 \text{Cons}_n + 0.10 \text{Impr}_n \in [0, 1] \quad (22)$$

D. Score Band Generation

$$\text{Expected} = 40 + 50 \cdot \text{SB} \quad (23)$$

$$\text{Min} = \text{Expected} - 10(1 - \text{SB}) \quad (24)$$

$$\text{Max} = \text{Expected} + 10 \cdot \text{SB} \quad (25)$$

A high SB produces a tight band (high certainty); a low SB widens it.

Algorithm 3 Teacher Intervention Recommendation

Require: Student set $\mathcal{S}$ for a course

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1: for all  $s \in \mathcal{S}$  do
2:   Compute Risk( $s$ ) via Eq. (27)
3: end for
4:  $\mathcal{S}_{\text{risk}} \leftarrow \{s : \text{Risk}(s) \geq 0.70\}$ 
5: for all  $s \in \mathcal{S}_{\text{risk}}$  do
6:    $s_{\text{weak}} \leftarrow \arg \max_{s_i \in W(s)} \text{sev}(s_i) \quad \triangleright \text{Eq. (28)}$ 
7:   Recommend: publish 5-question assignment on  $s_{\text{weak}}$ 
8:   Check  $\mathcal{Q}_{\text{ans}}(s)$  for recent remediation on  $s_{\text{weak}}$ 
9:   Output follow-up date: today + 2 days
10: end for

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E. Prediction Confidence

$$\text{Conf} = 0.5 + 0.4 \cdot \text{SB} + 0.1 \cdot \text{CompRate} \quad (26)$$

where CompRate is the fraction of all GATE DA questions answered.

F. Worked Example

Given: $\text{acc} = 78\%$, $\text{ELO} = 2350$, subject accuracies $[85, 72, 70, 75, 80]\%$ ($\bar{a} = 76.4\%$, $\sigma = 6.0\%$), $\Delta\text{acc} = +7\%$, $\text{CompRate} = 0.70$:

$$\text{Acc}_n = \min(1, 0.28/0.35) = 0.800$$

$$\text{ELO}_n = 750/1200 = 0.625$$

$$\text{Cons}_n = 1 - 0.060/0.774 = 0.922$$

$$\text{Impr}_n = 0.22/0.30 = 0.733$$

$$\text{SB} = 0.320 + 0.188 + 0.184 + 0.073 = 0.765$$

$$\text{Expected} = 40 + 50(0.765) = 78.2 \approx 78$$

$$\text{Band} = [76, 86], \quad \text{Conf} = 0.876$$

VI. RISK SCORING AND INTERVENTION ENGINE

A. Composite Risk Score

For each enrolled student s :

$$\text{Risk}(s) = 0.50(1 - \text{acc}(s)) + 0.30(1 - \text{comp}(s)) + 0.20\mathbb{I}[\text{ELO}(s) < 1800] \quad (27)$$

Students with $\text{Risk}(s) \geq 0.70$ enter the intervention queue.

B. Weakness Severity Ranking

Weak subjects in W are ranked for targeted assignment generation:

$$\text{sev}(s_i) = (1 - \text{acc}(s_i)) \cdot w_{\text{subj}}(s_i) \quad (28)$$

where w_{subj} encodes GATE DA subject weight (Probability and Statistics weight highest).

C. Intervention Algorithm

VII. COACH ENGINE LOGIC

A. Input State

The coaching engine receives:

$$\mathbf{c} = (\text{ELO}, \text{tier}, \text{acc}, W, S, k_{\text{streak}}, \mathcal{H}, \text{focus}) \quad (29)$$

where S is the strong-subject set and $\mathcal{H}$ is recent test history.

B. Proficiency Classification

$$\text{prof} = \begin{cases} \text{Low} & \text{ELO} < 1800 \\ \text{Medium} & 1800 \leq \text{ELO} \leq 2200 \\ \text{High} & \text{ELO} > 2200 \end{cases} \quad (30)$$

C. Deterministic Recommendation Rules

$$\text{advice} = \begin{cases} \text{Focus on fundamentals} & \text{prof} = \text{Low}, \text{acc} < 0.50 \\ \text{Tackle harder questions} & \text{prof} = \text{Medium}, \text{acc improving} \\ \text{Speed + mock practice} & \text{prof} = \text{High}, \text{acc} > 0.75 \\ \text{Topic review, lower diff} & \text{declining trend detected} \end{cases} \quad (31)$$

D. 7-Day Study Plan Allocation

Effort is distributed by weakness severity (Eq. (28)):

$$\text{effort}(d) = \begin{cases} 40\% & d \in \{1, 2\} : \text{highest-severity weak topic} \\ 30\% & d = 3 : \text{second-severity topic + foundations} \\ 20\% & d \in \{4, 5\} : \text{integrated multi-topic practice} \\ ? & d = 6 : \text{full mock test or timed paper} \\ 10\% & d = 7 : \text{revision of mock mistakes} \end{cases} \quad (32)$$